Rising Success, Persistent Gaps: Student-Level Statistics Outcomes Across California's Community College Placement-Reform Era, 2018 to 2025

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Abstract

California's Assembly Bills 705 (2017) and 1705 (2022) sharply limited remedial mathematics prerequisites and moved most community college students directly into transfer-level courses such as statistics. Critics predicted falling success. This practitioner case study follows official records for 867 enrollments in one instructor's transfer-level statistics sections at a large California community college from Fall 2018 to Fall 2025. In Fall and Spring terms, 42% of students passed before AB 705 and 65% under AB 1705, about one more student in every four. Across all terms, withdrawals fell from about one in five to one in forty. The yearly rise held in every check. Observed success rose in every racial and ethnic group, clearly for Latinx (44% to 61%) and Black students (27% to 62%, few students). The White and Latinx gap went from 25 to 24 points, but only a narrowing larger than the whole gap could have been detected, and race is missing for 38% of AB 1705 enrollments. Direct placement was followed by rising success, not the decline critics predicted. Whether the racial gap narrowed, these data cannot tell. Equity needs its own goals, tracked by group with complete records.

Keywords: community college, placement reform, AB 705, AB 1705, introductory statistics, equity gaps, corequisite support

1 Introduction

For many community college students, one quantitative course stands between them and transfer to a four-year university. More and more often, that course is introductory statistics. In Fall 2015, about 280,000 students took it in the mathematics programs of public two-year colleges, more than twice as many as five years earlier (Blair et al., 2018).

For decades, many students never reached that course. A placement test sent them first

to one or more remedial courses, and most did not come out the other end (Rodriguez et al., 2018). California changed this with two laws. Assembly Bill 705 (2017) required colleges to maximize the chance that a student enters and completes transfer-level English and mathematics within one year, and barred colleges from requiring remedial coursework unless a student was highly unlikely to succeed in transfer-level coursework (Assembly Bill 705, 2017); colleges were expected to comply by Fall 2019 (California Community Colleges Chancellor's Office, 2017). Assembly Bill 1705 (2022) went further. By July 1, 2023, colleges must directly place, and enroll in transfer-level English and mathematics, students who graduated from a United States high school or hold an equivalency certificate and plan to pursue a certificate, degree, or transfer. A college may require a remedial course only if its local research shows that the student is highly unlikely to succeed in the transfer-level course and the remedial course would improve the student's chance of completing transfer-level coursework within one year. The law also calls for support, such as corequisite courses, for students in transfer-level courses (Assembly Bill 1705, 2022).

The reforms remain contested. Critics argue that removing remedial prerequisites sends underprepared students into courses they will fail, a concern raised about statistics (Weissman, 2022) and, under AB 1705, about calculus (Dipierro, 2024). Supporters point to evidence that remedial sequences kept many students from ever reaching a transfer-level course. Statewide studies report that the share of first-time students taking transfer-level mathematics rose sharply after AB 705 (Cuellar Mejia et al., 2020), and that one-term completion of transfer-level mathematics among first-time mathematics students rose from 24% to 51% between Fall 2018 and Fall 2022 while racial and ethnic gaps remained (Cuellar Mejia et al., 2023). Statewide numbers, however, mix many courses, colleges, and instructors. They say less about what happened inside a single course, term after term, for each group of students.

This paper follows one transfer-level statistics course through the whole reform era and asks two plain questions.

1. **When students were placed directly into statistics, did a larger or a smaller share of them pass?**

2. **When AB 1705 came on top of AB 705, did success change for each racial and ethnic group, and did the gaps between groups get smaller?**

The data are the official records of every section one instructor, the author, taught from Fall 2018 to Fall 2025. The study is a practitioner case study: the kind of term-by-term, group-by-group review of one course that Bensimon (2007) argues faculty are well placed to do, carried out once and in full across both laws. Holding the instructor constant removes one source of change, a change in who teaches the course. It adds another: the instructor's

teaching, assessment, and grading may have changed over the same eight years, and the instructor assigned the grades. The design cannot separate the reforms from those changes, so the paper describes what happened, not what caused it. The paper adds three things to the statewide picture: student-level records for one course across both laws, success and racial gaps reported side by side under an explicit definition of equity, and a plain account of which findings hold up and which the data cannot settle.

2 Background and Conceptual Framework

2.1 Placement as a gate

For decades, placement tests sorted entering community college students into remedial or college-level courses. Clark (1960) described the community college's "cooling-out" function: students with bachelor's degree aspirations were gradually redirected toward lesser goals, and remedial placement was one of the ways this happened. Later evidence showed how often placement tests misjudged students. Scott-Clayton et al. (2014) found severe misassignment to be common, and far more often in one direction: students were placed in remediation they did not need much more often than in college-level courses they could not pass. In California before the reforms, most students placed in remedial courses never completed a transfer-level course (Rodriguez et al., 2018). Remedial placement mostly diverted students into remedial courses in place of college-level ones, with little evidence that it built their skills (Scott-Clayton & Rodriguez, 2015).

2.2 What the reforms put in its place

Two kinds of evidence shaped the reforms. The first was about placement itself. Among measures of high school achievement, high school grade point average was the most consistently useful predictor of how students would do in college courses (Bahr et al., 2019). The second was about what happens when students skip remediation. Accelerated pathways raised completion of transfer-level courses across 16 California colleges (Hayward & Willett, 2014). In a randomized trial, students assessed as needing remedial algebra who were placed instead in college-level statistics with support passed at higher rates than similar students placed in remedial algebra (Logue et al., 2016), and more of them later graduated (Logue et al., 2019). Corequisite remediation, in which support is taken alongside the college-level course, raised first-year passing of college-level mathematics in a statewide reform in Tennessee, though without clear long-term effects (Ran & Lin, 2022). A pathway course built

around statistics raised the share of students earning college mathematics credit (Yamada & Bryk, 2016).

California's early results pointed the same way. The share of first-time mathematics students who took a transfer-level course rose from 21% in Fall 2015 to 78% in Fall 2019, the first AB 705 term (Cuellar Mejia et al., 2020). Statewide, among students whose first mathematics course was transfer-level statistics or liberal arts mathematics, the share who passed within a year fell by 9 percentage points from Fall 2018 to Fall 2019 (68% to 59%), while the number who passed rose by 68% (Brohawn et al., 2021). Among first-time mathematics students who started in a transfer-level course, success then rose from 51% in 2019 to 57% in 2020 (Cuellar Mejia et al., 2021). By Fall 2024, 58% of first-time mathematics students completed a transfer-level course on their first attempt, and from Fall 2022 the gains were much larger for Latino and Black students than for Asian and White students (Cuellar Mejia & Wigul, 2026).

2.3 Two views of what should happen

These accounts lead to opposite predictions for the first question.

- **The preparation view.** Remedial prerequisites made sure students arrived ready. Removing them brings in less-prepared students, so course success should fall. This is the view of the faculty critics described in the Introduction (Dipierro, 2024; Weissman, 2022).

- **The gatekeeping view.** Placement tests and remedial sequences filtered out many students who could have passed. Removing the gate lets more of them in without lowering success, and success may rise as support moves inside the course.

They also lead to two expectations for the second question.

- **Narrowing.** If misplacement into remediation fell more heavily on Latinx and Black students, removing the gate should help them most, and their gaps to White students should narrow.

- **A rising tide.** A rule applied the same way to everyone may lift every group by similar amounts and leave the gaps where they were.

A case study of one course cannot settle which view is right. It can show whether the decline the preparation view predicts appeared in these sections, and whether the gaps moved. Table 5, in the Discussion, sets each expectation beside what we found.

2.4 What we mean by equity

Equal access is not the same as equitable outcomes. Bensimon (2007) argued that research on student success too often treats unequal outcomes as a property of students rather than of institutions and practice; an equity lens asks whether outcomes, not only opportunities, are fair across groups. Because "equity" can mean several things, we separate three criteria and report each one.

1. **Improvement within a group:** the group's success rate rose from before AB 1705 to under AB 1705.
2. **A narrowing gap:** the difference between the group's success rate and that of White students, the highest-success large group in these data, became smaller.
3. **Parity:** the gap reached zero.

In this paper, *equity* means the second and third criteria. A group can improve (criterion 1) while its gap stays the same, if the reference group improves just as much. Keeping these apart prevents a rising tide from being read as a closing gap. White students are the reference because they had the highest success rate of the four groups compared, both before and under AB 1705. Other references are possible. Gaps can also be measured against the rate for all students or for all other students, as in California's student equity planning (California Community Colleges Chancellor's Office, 2022); Appendix Table A3 gives the gaps both ways, and the conclusions do not change.

2.5 Why statistics

Introductory statistics suits this question. Under AB 705, about two-thirds of first-time students who enrolled directly in transfer-level mathematics took statistics or liberal arts mathematics (Cuellar Mejia et al., 2020), and statistics instruction emphasizes statistical thinking and conceptual understanding, and most introductory courses require no more than high school algebra (GAISE College Report ASA Revision Committee, 2016). Little longitudinal, student-level evidence follows both overall success and the gaps between groups in one statistics setting across the whole reform era. This study provides one such case.

3 Data and Methods

3.1 Setting and sample

The data are official student records released by the Office of Institutional Research of Santa Monica College for every transfer-level introductory statistics section (MATH 54, renamed STAT C1000 in Fall 2025 under California's common course numbering) taught by one instructor. It is a large college, the only one in its district, in urban Los Angeles County, with about 23,000 students in Fall 2024, about 40% of them Latinx (National Center for Education Statistics, n.d.). The sample is 867 enrollments across 17 terms, Fall 2018 through Fall 2025. Terms in which the instructor taught no section, such as Spring 2023, Spring 2024, and Spring 2025, are absent. The enrollments belong to 852 distinct students (15 repeat enrollments, 1.7%), so enrollment-level and student-level results are effectively the same. The release includes student identifiers and final grades; the analytic file keeps neither, and the counts that need them (distinct students, grades, and corequisite co-enrollment) were computed from the release and are reported only as term totals. The records do not include delivery mode or any measure of prior preparation.

The support course. The same instructor also taught a corequisite support course (MATH 54C), graded pass or no pass and taken alongside the statistics course. It was first offered in Fall 2019, the first AB 705 term, and was designed for students who would have been placed below transfer level before AB 705 (Santa Monica College Mathematics Department, 2024). In terms when it was offered, between 38.5% and 100% of the course's students were also enrolled in it. Enrollments in the support course itself are not counted, so that no student is counted twice.

3.2 What we measured

Outcome (dependent variable). *Course success*, the college's own success flag: a final grade of A, B, C, or P (pass) counts as success; a D, F, NP (no pass), W (withdrawal), or incomplete does not. A C is the threshold for transfer credit. Pass and no-pass grades are rare in this course (4 and 12 of 867 enrollments). The release contains no excused withdrawal (EW) grades (0 of 867), so every withdrawal in the records is a W and counts against success. The records do not show whether any excused withdrawals were left out of the release, or at what point in the term enrollment was counted; Finding 2 returns to this.

What success is compared across (independent variables).

- *Time*: years since Fall 2018, with each term placed where it falls in the calendar year, so that a five-week Winter term is not counted as a full semester.

- *Policy period*: Pre-AB 705 (Fall 2018 and Spring 2019); AB 705 + COVID-19 (Fall 2019 to Spring 2021); Post-AB 705 (Fall 2021 to Winter 2023); and AB 1705 (Fall 2023 to Fall 2025). The cut points are the statewide dates, Fall 2019 for AB 705 and July 1, 2023 for AB 1705. The AB 705 date matches the college's own timeline: its Office of Institutional Research reports that the college implemented AB 705 for mathematics in Fall 2019, replacing placement by examination with placement based on high school records and adding transfer-level courses with corequisite support (Lawler, 2020). Below-transfer mathematics courses remained available until they were last offered in Summer 2022 (Santa Monica College Mathematics Department, 2024), so direct placement was phased in rather than switched on at once. With those courses gone, the AB 1705 date marks the statewide rule rather than a known local change.

- *Term type*: regular (Fall or Spring) or intersession (Winter or Summer). Intersession terms appear in these data only from 2022 and pass at higher rates, so the main analysis accounts for them.

- *Race and ethnicity*, as officially recorded: White, Latinx (recorded as "LatinX"), Black, Asian, and Other or unknown (multiracial, Pacific Islander, Native American, and "unknown").

- *Gender*, as officially recorded, used descriptively.

3.3 The COVID-19 terms

Three terms, Spring 2020, Fall 2020, and Spring 2021, were taught under emergency remote instruction and emergency grading rules that changed how withdrawals and pass or no-pass grades were recorded. Spring 2020 and Spring 2021 record no withdrawals at all, and Spring 2020 shows an unusually high success rate (76.9%). Because grades in these terms may not mean what grades meant in other terms, the main trend analysis leaves them out, and the main trend results are repeated with them included and with the window extended through Spring 2022. These terms fall inside the AB 705 period, so the AB 705 + COVID-19 period mixes the first years of reform with the pandemic, and the data cannot separate the two. The group comparisons keep the COVID-19 terms, because removing them would leave some groups very small, and are repeated without them. Appendix Table A1 repeats the main results with and without them, and the online supplement gives the details.

3.4 Missing data

Course success is recorded for every enrollment. Race and gender are missing for three whole terms (Summer 2022, Summer 2025, and Fall 2025), 116 enrollments (13.4%). These enrollments are kept in every analysis of overall success. The group comparisons use the enrollments with race recorded. The omission matters: enrollments without demographics passed at 71.6%, against 55.4% for the rest, and most fall in the AB 1705 period, where race is recorded for only 173 of 277 enrollments. A sensitivity analysis asks how much this could change the equity finding.

3.5 How we analyzed the data

Question 1. We report the share of students who passed in each period and term, and fit logistic regressions of course success: the standard model for a yes-or-no outcome. The main model (Model 2) asks how the odds of passing changed with each calendar year, accounting for whether a term was regular or intersession. Model 1 uses time alone, Model 3 adds race and ethnicity, and Model 4 compares the four policy periods directly. In Models 1 to 3, uncertainty is computed allowing students in the same term to be alike (standard errors clustered by term). With only 14 terms, clustered standard errors can be too small, so we also checked the trend with a wild cluster bootstrap (Model 2), and with a bootstrap that resamples whole terms and a model of term totals that allows for swings from term to term (both Model 1). We report each result as an odds ratio with its 95% confidence interval (CI), the range of values consistent with the data, and, to show what the odds mean, as the predicted share of students passing at the start and end of the study and the average change in percentage points per year. When the 95% CI includes no change, the data do not show a clear change. We also tested whether success jumped when either law took effect (online supplement).

Question 2. For each racial and ethnic group, we compare success before AB 1705 (Fall 2018 to Winter 2023) with success under AB 1705 (Fall 2023 to Fall 2025). We use the AB 1705 date rather than the AB 705 date because the period before AB 705 has only 85 enrollments, 23 of them White students, too few for group comparisons. This comparison therefore measures what changed with AB 1705 on top of AB 705. For each group we report the change in success in percentage points (criterion 1), its gap to White students before and under AB 1705, and the change in that gap (criterion 2), each with a 95% CI. We also report how large a narrowing these group sizes could reliably detect. For criterion 3, a gap is *established* when its whole 95% CI lies above zero, and parity would be established only if the whole interval lay within 5 points of zero; this margin is a judgment, not a published

standard, and no conclusion depends on it. The online supplement describes the interval methods and two further models of the gaps.

Software and AI use. All analyses were run in Python 3.13.2 with pandas 3.0.3, NumPy 2.2.5, SciPy 1.17.1, statsmodels 0.14.6, and Matplotlib 3.10.3. One script produces every number, and every table and figure except Table 4, from the data; a second script turns those results into the rounded values and Table 4 used in the text and checks the plain statements against them; a third, private script turns the identified release into the term totals the first script uses. College-wide, statewide, and enrollment figures from other sources are typed from the cited documents. Generative AI (Claude, Anthropic) was used to assist with writing, running, and checking the analysis code, checking references, and drafting, editing, and proofreading the text. The author designed the study, directed and reviewed all of this work, and takes full responsibility for the content.

3.6 Ethics

The author asked the college's institutional review board to review the study. At the time, the college was not conducting full institutional review board reviews because of budget and staffing limits, and it accepted only approvals obtained through other organizations. Its Office of Institutional Research advised that the study could proceed as educational research with the approval of the department chair. The chair approved, and the office then released the records, without names. No approval or exemption determination was obtained from any review board; departmental approval is not ethics review, and we do not present it as such. The study is a secondary analysis of existing administrative records, with no research intervention and no interaction with students for research purposes; no consent was sought, because no student was contacted. Student identifiers in the release were used only to count distinct students and corequisite co-enrollment, and the analytic file contains none. Some small cells could be re-identifiable, so the public deposit will be coarsened (see Statements). The author was also the instructor of record for these sections and assigned the grades analyzed here; this is disclosed in the Statements and treated as a limitation.

4 Results

4.1 Who enrolled

Table 1 describes the students in each period. Enrollment per term rose from 42.5 before AB 705 to 67.8 during AB 705 and COVID-19, fell to 39.0 after AB 705, and was 55.4 under AB 1705; these counts also reflect how many sections the instructor was assigned. The course

Table 1: Students and course success by policy period.

	Pre-AB 705	AB 705 + COVID-19	Post-AB 705	AB 1705
Terms	2	4	6	5
Enrollments	85	271	234	277
per term	42.5	67.8	39.0	55.4
Course success, %	42.4	49.4	59.0	69.0
Fall and Spring terms only, %	42.4	49.4	54.7	65.4
Withdrew (W), %	18.8	11.4	11.1	2.5
Passed, among those who did not withdraw, %	52.2	55.8	66.3	70.7
Co-enrolled in the corequisite support course, %	0.0	44.3	62.0	43.3
Enrollments with race recorded	85	271	222	173
White, %	27.1	23.2	23.0	18.5
Asian, %	15.3	8.5	4.5	6.4
Latinx, %	36.5	48.3	56.3	56.1
Black, %	10.6	8.9	7.2	7.5
Other or unknown, %	10.6	11.1	9.0	11.6
Course success by group, % (n)				
White	69.6 (23)	61.9 (63)	76.5 (51)	84.4 (32)
Asian	38.5 (13)	78.3 (23)	50.0 (10)	72.7 (11)
Latinx	32.3 (31)	37.4 (131)	52.8 (125)	60.8 (97)
Black	11.1 (9)	29.2 (24)	31.2 (16)	61.5 (13)
Other or unknown	44.4 (9)	70.0 (30)	65.0 (20)	80.0 (20)
Gap, White minus Latinx, pp	37.3	24.5	23.7	23.6
Female, % of women and men	61.9	64.6	55.9	64.9
Course success, women, %	46.2	49.7	55.3	68.5
Course success, men, %	34.4	49.0	60.8	66.7

pp: percentage points. Race and gender percentages and success by group are among enrollments with the field recorded. Race and gender are missing for Summer 2022, Summer 2025, and Fall 2025, so the AB 1705 composition rests on 173 of 277 enrollments. "Other or unknown" combines multiracial, Pacific Islander, Native American, and recorded-unknown students. Gender "Other" ($n = 5$) is not shown.

became more Latinx: among enrollments with race recorded, the Latinx share rose from 36.5% before AB 705 to 56.1% under AB 1705, while the shares of White (27.1% to 18.5%), Asian (15.3% to 6.4%), and Black students (10.6% to 7.5%) fell. Women were a majority in every period.

4.2 Finding 1: a larger share of students passed, not a smaller one

Before AB 705, 42 of every 100 students passed the course. Under AB 1705, 65 of every 100 passed in Fall and Spring terms, and 69 of every 100 counting the short Winter and Summer terms (Table 1; Figure 1). Comparing Fall and Spring terms only, the rise was about 23 percentage points, or roughly one more student passing out of every four who enrolled. Success rose step by step through the periods: 42.4% before AB 705, 49.4% during AB 705 and COVID-19, 59.0% after AB 705, and 69.0% under AB 1705.

The decline critics predicted did not appear, with one possible exception. Fall 2019, the first term under AB 705 and the largest term, was the lowest of all 17 terms (32.1% of 84 students passed). It was not clearly different from the rate before AB 705 (Fisher's exact test, $p = .204$), and success rose in the terms that followed.

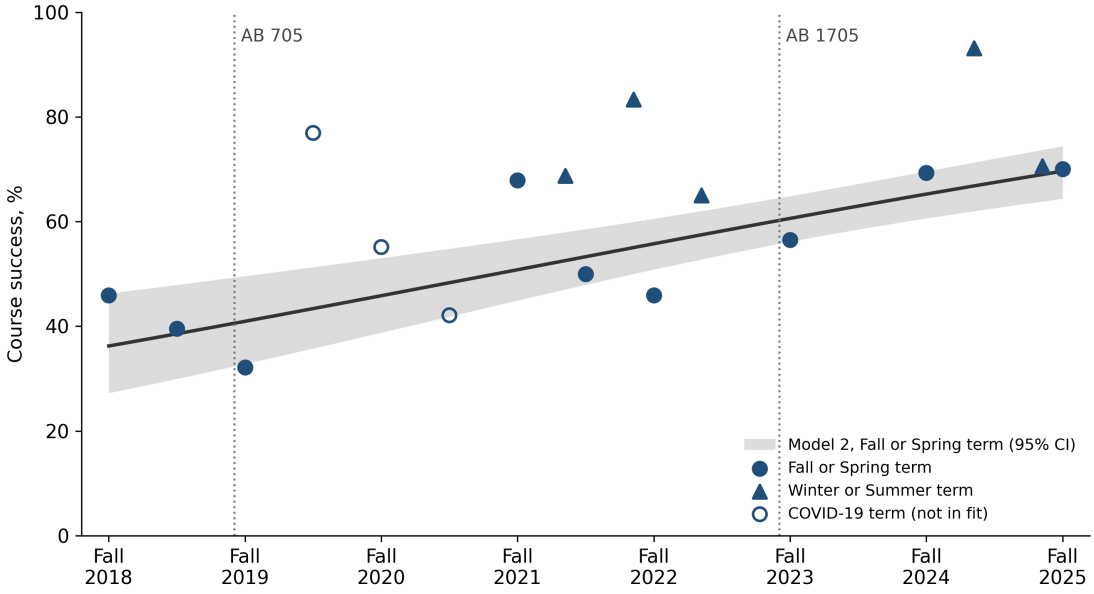


Figure 1: Course success by term, Fall 2018 to Fall 2025, placed on the calendar. Circles are Fall and Spring terms, triangles are Winter and Summer terms, and open circles are the three COVID-19 terms, which are not in the fitted model. The line and band are Model 2's predicted success for a regular term with its 95% confidence interval. Dotted lines mark the implementation dates of AB 705 (Fall 2019) and AB 1705 (Fall 2023).

4.3 Finding 2: fewer students withdrew, and more of those who stayed passed

Two things moved together. Before AB 705, about one student in five withdrew (18.8%); under AB 1705, about one in forty did (2.5%). Among students who stayed in the course, the share who passed rose from 52% to 71% (Table 1). In Fall and Spring terms alone, 1.4% withdrew under AB 1705 and 66.4% of those who stayed passed. The records cannot say why withdrawals fell; changes in withdrawal deadlines, in how students used them, or in the instructor's practice are all possible. One recording question matters here. Statewide success rates leave excused withdrawals (EW), granted for extenuating circumstances, out of the count of enrollments (California Community Colleges Chancellor's Office, n.d.). During the pandemic an executive order eased the rules for excused withdrawals, and a permanent change to the state regulations, expected to begin in Fall 2022, treats every withdrawal for

extenuating circumstances as an excused withdrawal and no longer requires documentation from the student (California Community Colleges Chancellor's Office, 2023). That change covers the whole AB 1705 period. The release contains no EW grades, and five terms (Spring 2020, Spring 2021, Fall 2021, Summer 2022, and Winter 2025) record no withdrawals at all. If excused withdrawals were left out of the release, some students who withdrew are missing from these counts, and part of the fall in withdrawals, and so part of the rise in success, would reflect how withdrawals were recorded rather than what students did. The records cannot settle this.

4.4 Finding 3: the rise was steady, and it held up

Table 2 gives the full results of all four models. In the main model (Model 2), the odds of passing were 1.22 times as high with each calendar year, about 22% higher each year (95% CI 1.13 to 1.32). In plain terms, a student in a regular term had a predicted 36% chance of passing in Fall 2018 (95% CI 27.3% to 46.3%) and a 70% chance in Fall 2025 (95% CI 64.4% to 74.4%), about 4.8 percentage points a year from start to end. Averaged over the students in the sample, the rise was 4.5 points a year (the average marginal effect; 95% CI 3.0 to 6.0); it is smaller mainly because success in the intersession terms was already high, where the same change in odds moves fewer percentage points. The predicted starting point sits below the observed Fall 2018 rate (45.9%) partly because the low Fall 2019 term pulls the line down; the observed Fall rate rose from 45.9% in Fall 2018 to 70.0% in Fall 2025.

The rise was not explained by the changing makeup of the class. Intersession terms passed at higher rates (odds ratio 1.99), but the trend holds with them accounted for. Adding race and ethnicity leaves it in place and slightly larger (Model 3: 1.27 per year), so the rise does not come from changes in the racial and ethnic composition of enrollment. Other changes in who enrolled, such as prior preparation, are not in the records.

The rise was a steady climb, not a jump when a law took effect. Comparing whole periods is less certain. Once the analysis allows for swings from term to term, the difference between the two terms before AB 705 and the AB 1705 period is borderline ($p = .053$), although the odds of passing under AB 1705 were 2.61 times as high (Model 4, conventional 95% CI 1.57 to 4.33). A model that looked for a jump at each law's start date found no clear jump at either date (Table 4, Panel A).

Table 4, Panel A, shows the trend under eight versions of the analysis: with the COVID-19 terms included or the window extended, with race added, and with three different ways of measuring uncertainty. In every version, success kept rising. With the COVID-19 terms included, the trend is slightly smaller (1.17 per year) but still clearly positive. The wild

cluster bootstrap, built for analyses with few clusters, also rejects no trend for Model 2 ($p = .019$, all 16,384 sign patterns of the 14 terms).

Table 2: Logistic regressions of course success: full results. Odds ratios (p), with 95% confidence intervals below.

	Model 1	Model 2	Model 3	Model 4
Year (per calendar year)	1.26 (< .001) [1.15, 1.37]	1.22 (< .001) [1.13, 1.32]	1.27 (< .001) [1.16, 1.40]	
Intersession term (vs Fall or Spring)		1.99 (.008) [1.19, 3.31]	2.12 (.004) [1.26, 3.55]	2.08 (.002) [1.32, 3.29]
AB 705 + COVID-19 (vs Pre-AB 705)				1.33 (.254) [0.81, 2.18]
Post-AB 705 (vs Pre-AB 705)				1.62 (.067) [0.97, 2.71]
AB 1705 (vs Pre-AB 705)				2.61 (< .001) [1.57, 4.33]
Latinx (vs White)			0.31 (< .001) [0.21, 0.47]	
Black (vs White)			0.14 (< .001) [0.07, 0.28]	
Asian (vs White)			0.51 (.165) [0.20, 1.32]	
Other or unknown (vs White)			0.57 (.027) [0.35, 0.94]	
Race not recorded[a]			0.32 (< .001) [0.19, 0.53]	
Intercept (odds)	0.57 (.012) [0.37, 0.88]	0.57 (.008) [0.37, 0.86]	1.20 (.569) [0.64, 2.28]	0.73 (.160) [0.48, 1.13]
Enrollments	680	680	680	867
Terms	14	14	14	17
COVID-19 terms	excluded	excluded	excluded	included
Standard errors	by term	by term	by term	conventional
McFadden pseudo R^2	0.045	0.055	0.095	0.035
AIC	888.8	881.5	854.4	1150.9

Outcome: course success (a grade of A, B, C, or P = 1). Year is calendar years since Fall 2018. Model 1: time only. Model 2 (main model): adds term type. Model 3: adds race and ethnicity (reference: White). Model 4: the four policy periods (reference: Pre-AB 705, two terms) and term type, all terms. Models 1 to 3 use standard errors clustered by term; Model 4 uses conventional standard errors because its reference period has only two terms (Appendix Table A2 gives the alternatives). An odds ratio above 1 means higher odds of passing. AIC can be compared among Models 1 to 3, which use the same 680 enrollments, but not with Model 4, which uses all 867. [a] Race is missing for three whole terms (Summer 2022, Summer 2025, Fall 2025); this indicator marks those enrollments and is not a group effect.

4.5 Finding 4: observed success rose in every group, clearly for Latinx and Black students

Table 3 and Figure 2 compare each group before and under AB 1705. Every group's success rate rose. For Latinx students, the largest group, it rose from 44% to 61%, about 17 percentage points (95% CI 5.8 to 28.0). For Black students it rose from 27% to 62%, about 35 points (95% CI 5.6 to 58.2), although only 13 Black students enrolled under AB 1705. Both increases are clear. For White students (69% to 84%) and Asian students (61% to 73%), the rises were not clearly different from no change. By the first equity criterion, then, Latinx and Black students' success improved.

Table 3: Success by race and ethnicity before and under AB 1705, and the gap to White students.

	Success rate, % (n)		Change within group		Gap to White students, pp [95% CI]		
Group	Pre-AB 1705	AB 1705	pp [95% CI]	p	Pre-AB 1705	AB 1705	Change
White	68.6 (137)	84.4 (32)	15.8 [−1.9, 27.8]	.085	reference	reference	reference
Asian	60.9 (46)	72.7 (11)	11.9 [−20.1, 34.5]	.729	7.7 [−7.4, 23.8]	11.6 [−12.2, 42.2]	3.9 [−29.4, 37.2]
Latinx	43.6 (287)	60.8 (97)	17.3 [5.8, 28.0]	.003	25.1 [15.0, 34.2]	23.6 [5.0, 36.8]	−1.5 [−20.1, 17.1]
Black	26.5 (49)	61.5 (13)	35.0 [5.6, 58.2]	.025	42.1 [26.1, 54.7]	22.8 [−3.4, 50.3]	−19.2 [−52.0, 13.5]

pp: percentage points. Pre-AB 1705: Fall 2018 to Winter 2023, including the COVID-19 terms. Change within group: AB 1705 minus Pre-AB 1705, with 95% Newcombe confidence intervals and Fisher's exact test. Gap: White success rate minus the group's rate, with 95% Newcombe intervals; change in gap: AB 1705 gap minus Pre-AB 1705 gap, with a 95% Wald interval. Enrollments with race recorded only; race is missing for 104 of 277 AB 1705 enrollments.

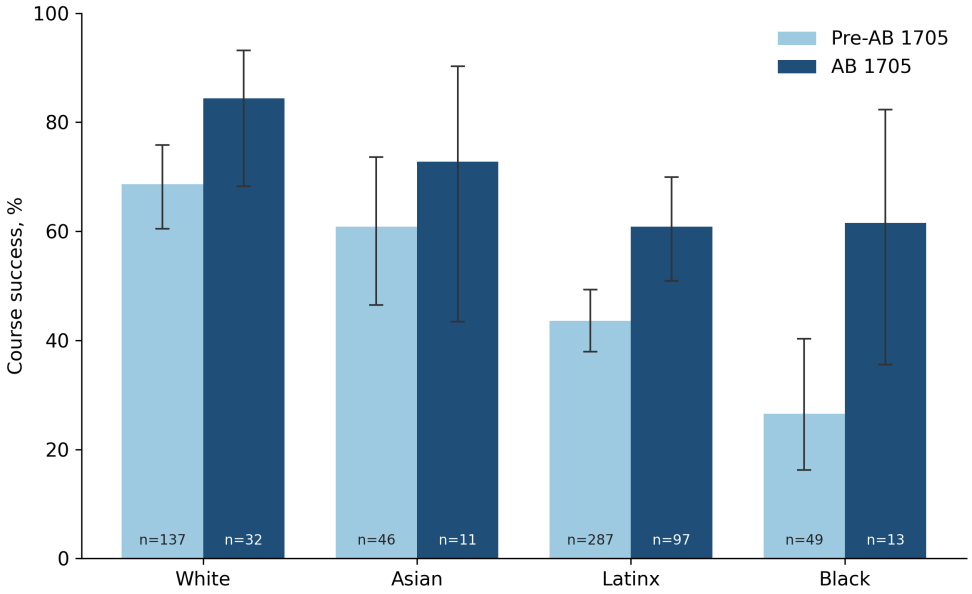


Figure 2: Course success by race and ethnicity before and under AB 1705, with 95% Wilson confidence intervals. The number of students, n, is shown at the base of each bar. Overlapping intervals are not a test of difference; the direct tests are in Table 3.

4.6 Finding 5: the White and Latinx gap barely moved, and only a narrowing larger than the whole gap was detectable

Because White students' success rose too, the point estimates of the gaps barely moved. The gap between White and Latinx students has the most data behind it. Before AB 1705, the White success rate was 25.1 percentage points higher than the Latinx rate (95% CI 15.0 to 34.2); under AB 1705, it was 23.6 points higher (95% CI 5.0 to 36.8). The gap barely moved in any period from Fall 2019 on: about 24 points in each (Table 1). Before AB 705 the gap was 37.3 points, but with 23 White and 31 Latinx students in those two terms, the change at that boundary (−12.8 points, 95% CI −41.7 to 16.1) cannot be assessed.

By the second criterion, then, the study cannot give an answer. With these group sizes, only a narrowing of about 27 points or more could have been reliably detected (with 80% power), more than the whole 25-point gap. The change in the gap, −1.5 points, has a 95% CI from −20.1 to 17.1: it fits a gap that stayed where it was, and it fits a large narrowing. Table 4, Panel B, shows the same answer without the COVID-19 terms, in Fall and Spring terms only, and in a model that follows each group's success over every term.

The gap for Black students fell from 42.1 to 22.8 points, but with only 13 Black students under AB 1705 the change is very imprecise (95% CI −52.0 to 13.5). The Asian gap was imprecise in both blocks ($n = 11$ under AB 1705). By the third criterion, a gap to White

students is established for Latinx students in both blocks and for Black students before AB 1705. No comparison is precise enough to establish parity.

Could the missing race data change this? Under AB 1705, race is missing for 104 of 277 enrollments (38%), and those students passed at 70.2%. Race is missing for whole terms, not for scattered students, so whether it is missing depends on the term, not on anything about the student; what is unknown is the racial makeup of those terms and how White and Latinx students did within them. We gave the missing enrollments the racial makeup of the AB 1705 students whose race was recorded, kept their overall pass rate as observed, and asked how differently White and Latinx students among them would have had to do for the AB 1705 gap to reach zero. The answer is that Latinx students would have had to pass about 39 points more often than White students. In no term with race recorded did Latinx students pass more than about 6 points more often than White students (online supplement). Even a reversal of 30 points would leave a gap of 3.4 points, a narrowing of about 22 points hidden in the missing records. So the missing records could hide a substantial narrowing, but a gap that closed completely would need a pattern far outside anything observed.

4.7 The support course

From Fall 2019 on, many students took the corequisite support course alongside the statistics course. Over all terms, 222 of the 385 students who took it passed the transfer-level course. In some terms every student took it, so a comparison is possible only in the 9 terms in which some but not all students took it. There, co-enrolled students passed somewhat less often than the others (53.1% and 59.5%), a difference not clearly distinguishable from zero. Who took support was not random, so neither figure estimates the effect of support.

4.8 Gender

Women and men passed at nearly identical rates overall (55.3% and 55.1%; difference 0.2 points, 95% CI −7.1 to 7.6).

5 Discussion

This study has three main findings. First, when students were placed directly into transfer-level statistics, a larger share of them passed, not a smaller one: 42% before AB 705 and 65% under AB 1705 in Fall and Spring terms, a steady yearly rise that held in every check. Second, withdrawals nearly disappeared, and a larger share of the students who stayed

Table 4: How the findings held up.

A. Did success keep rising over time? Version of the analysis	Odds ratio per year [95% CI]	Still rising?
Main result: adjusted for summer and winter terms, without the COVID-19 terms (Model 2)	1.22 [1.13, 1.32]	Yes
Time only (Model 1)	1.26 [1.15, 1.37]	Yes
With the COVID-19 terms included (Model 2)	1.17 [1.06, 1.29]	Yes
COVID-19 window extended through Spring 2022 (Model 2)	1.23 [1.14, 1.33]	Yes
Also adjusted for race and ethnicity (Model 3)	1.27 [1.16, 1.40]	Yes
Model 3, students with race recorded only	1.28 [1.18, 1.40]	Yes
Resampling whole terms (bootstrap, Model 1)	1.26 [1.14, 1.40]	Yes
Allowing extra swings between terms (term-level model, Model 1)	1.26 [1.13, 1.40]	Yes
Jump in success when AB 705 took effect (segmented model)	0.65 [0.25, 1.70]	No clear jump
Jump in success when AB 1705 took effect (segmented model)	0.87 [0.33, 2.30]	No clear jump
B. Did the White and Latinx gap narrow? Version of the analysis	Change in gap, pp [95% CI]	Narrowing shown?
Main result: before vs under AB 1705, all terms	−1.5 [−20.1, 17.1]	No
Without the COVID-19 terms	−0.6 [−20.4, 19.1]	No
Fall and Spring terms only	−0.2 [−21.4, 20.9]	No
Every term, each group with its own trend (Fall 2018 vs Fall 2024)	−8.8 [−32.4, 14.9]	No
Students with no race recorded under AB 1705 given every assumed White and Latinx difference from 30 points one way to 30 points the other	AB 1705 gap 3.4 to 26.0 pp	Gap stays above zero (point estimates)

Panel A: odds ratio for one calendar year with its 95% CI; "Still rising?" is Yes when the whole interval lies above 1. The segmented model fits one common trend plus a step at each law's start date; a step near 1 means no jump, and neither step differs clearly from no jump under any of three kinds of standard errors (Appendix Table A2). Panel B: change in the White minus Latinx gap in percentage points (negative means narrowing) with its 95% CI; "Narrowing shown?" is Yes only when the whole interval lies below zero. The last row is the sensitivity analysis for missing race described in the text. Details of every version are in the online supplement.

passed. Third, observed success rose in every racial and ethnic group, clearly for Latinx and Black students, while the gap between White and Latinx students barely moved; the study could not have detected a narrowing smaller than the whole gap. Rising success is not, by itself, equity.

5.1 What this means for policy and practice

The decline critics feared did not appear. In these sections, the drop in success that the preparation view predicted did not appear over the period as a whole. These data offer no support for restoring remedial prerequisites in statistics on the grounds that direct placement lowers success.

Access is not equity. A rule applied the same way to everyone was followed by higher observed success in every group, and the gap between White and Latinx students barely moved. Equity needs its own goals, set and tracked separately from overall success. A college that watches only its overall pass rate could report steady progress while its gaps stay where they were.

Track results by group, with complete records. Race was missing for three whole terms here, including the two most recent. Without complete demographic records, a college cannot know whether its gaps are closing. Recording race for every term, and reviewing results by group each term, is a low-cost first step.

Support inside the course is worth testing. The support course reached many students, and 222 students who took a course designed for students once placed below transfer level passed the transfer-level course. These data cannot show whether support raised success. Prior work points to support taken alongside the course (Logue et al., 2019; Ran & Lin, 2022), pathway course designs (Yamada & Bryk, 2016), and regular review of results by group by the faculty who teach the course (Bensimon, 2007) as candidates worth testing for closing gaps.

What would answer the question. The most useful next study would pool many instructors across several colleges over the same years, with placement and preparation data and groups large enough to detect a narrowing of a few points. That would separate the reforms from any one instructor's teaching and grading.

5.2 No sustained decline under direct placement

The preparation view predicted that success would fall once remedial prerequisites were removed. In these sections it did not fall over the period as a whole. This fits the gatekeeping view, although these data cannot show which view is right. It parallels statewide reports that

one-term completion of transfer-level mathematics among first-time mathematics students roughly doubled between Fall 2018 and Fall 2022 (Cuellar Mejia et al., 2023), a related but different measure, and experimental evidence that students placed directly in college-level statistics with support pass at higher rates than similar students sent to remediation (Logue et al., 2016). A larger share passing does not mean fewer students failed in number. When enrollment grows, the number who do not pass can rise too: in one large district's statistics courses, Fall 2019 brought 981 more Black and Latinx students who passed than Fall 2018, and 2,450 more who failed (Weissman, 2022).

Fall 2019, the first AB 705 term, had the most enrollments and the lowest success rate. It may be the short-lived drop the preparation view predicts, or ordinary variation from term to term, since the rate was not clearly different from the rate before AB 705. The dip was not peculiar to these sections, though it was deeper here. College-wide, success in the course fell from 51.4% in Fall 2018 to 45.6% in Fall 2019, as its enrollment grew from 1,649 to 2,349 (Lawler, 2020), and statewide, the one-year success rate of students who started in transfer-level statistics or liberal arts mathematics fell by 9 points over the same year (Brohawn et al., 2021).

The enrollment pattern is consistent with the gate opening. Enrollment per term rose by more than half in the AB 705 period, and the Latinx share of enrollment grew to more than half in the last two periods (56.3% and 56.1%). In Clark's (1960) terms, fewer students may have been cooled out before reaching the course. This is weak evidence, because enrollment in one instructor's sections also depends on how many sections the instructor was assigned.

The design cannot separate the reforms from several rival explanations: changes in the instructor's teaching, assessment, and grading; the near disappearance of withdrawals; differences between students who choose intersession terms; changes in delivery mode after the pandemic; and shifts in statewide grading after 2020. A hidden change could mask a negative policy effect as easily as create a positive one. The claim these data support is narrower and still useful: no sustained decline of the kind critics predicted appeared here.

5.3 Improvement is not equity

By the first equity criterion, Latinx and Black students' success improved. By the second, the study cannot give an answer. The White and Latinx gap barely moved because White students' success rose by a similar amount, though the White change is imprecise. That is closer to the rising-tide expectation than to narrowing, but only weakly: the interval for the change in the gap is wide, the missing race data could hide a narrowing, and a model that follows each group over every term leans toward a modest narrowing (-8.8 points) with an

Table 5: What each view expected, and what we found.

View	What it expected	What we found in these sections
Preparation	Success falls once remedial prerequisites are removed.	Not seen. Success rose from 42% to 65% in Fall and Spring terms; only the first AB 705 term dipped, and not clearly.
Gatekeeping	More students reach the course without lower success; support moves inside the course.	Consistent. Success rose steadily, and many students took the corequisite support course. The design cannot rule out other explanations.
Narrowing	Gaps to White students shrink as the gate opens.	Not shown, and not ruled out. The White and Latinx gap was 25 points, then 24; only a narrowing larger than the whole gap was detectable.
Rising tide	Every group rises by similar amounts; gaps stay.	Closest to the block comparison, not established. The White and Latinx gap barely moved there; the every-term model leans toward modest narrowing, and neither is precise enough to choose.

interval that also includes no change.

The gap is not peculiar to these sections. College-wide, White and Latinx students' success in the course differed by about 25 points in both Fall 2018 (66.3% and 41.6%) and Fall 2019 (61.1% and 36.4%) (Lawler, 2020); we found no college-wide figures by race for later terms. Statewide, the picture is partly different. Between Fall 2022 and Fall 2024, one-term completion of transfer-level mathematics among first-time mathematics students kept rising, and Latino and Black students' completion rose much more than Asian and White students' (Cuellar Mejia & Wigul, 2026). That measure counts only first-time students, while these sections include continuing and repeating students. Statewide success in transfer-level statistics taught in mathematics departments, among first-time mathematics students, was about 60% in 2024 (Public Policy Institute of California, 2026), against 69.3% in these sections in Fall 2024, though the two rates describe different groups of students.

Bensimon's (2007) argument explains why this matters. Placement reform changed the rules of access, the same for everyone. Changing who gets in is not the same as changing who succeeds, and the second needs attention of its own.

5.4 Strengths and limitations

Strengths include official student-level records for every section of one course across both laws, the same instructor throughout, an explicit definition of equity with each criterion reported separately, full model results, and scripts that produce every reported number, with every result checked several ways and the analyses that point another way reported in the supplement.

The study has six main limitations.

1. **One instructor, one college.** The trend cannot be separated from changes in the instructor's teaching, assessment, and grading, and the instructor assigned the grades that make up the outcome. Other changes over the same years, such as delivery mode, the sharp fall in withdrawals and how withdrawals were recorded, the course's renumbering, campus supports, and statewide grading patterns, are not measured. The results may not generalize, so we describe the rise without attributing it to the reforms.
2. **Success among those who enrolled.** The reforms are usually judged by throughput, the share of entering students who complete a transfer-level course. These data measure success among students who enrolled, a different quantity, and contain no measure of prior preparation.
3. **Small groups.** Under AB 1705, Black ($n = 13$), Asian ($n = 11$), and White ($n = 32$) students are few, so their rates, and every AB 1705 gap, are imprecise; only a narrowing of the White and Latinx gap larger than the whole gap could reliably have been detected.
4. **Missing demographics.** Race and gender are missing for three terms (13.4% of enrollments), mostly under AB 1705. The White and Latinx gap stays positive under every scenario examined, but the missing records could hide a narrowing of as much as about 22 points.
5. **Policy timing and the pandemic.** Below-transfer courses remained an option until Summer 2022, so direct placement was phased in; the baseline before AB 705 has only two terms; and emergency grading rules in the COVID-19 terms cannot be reconstructed, so results are reported both with and without those terms.
6. **A coarse outcome.** Passing with a C or better does not measure depth of learning or later success.

6 Conclusion

Across eight years of California's placement reforms, success in one instructor's transfer-level statistics sections rose, from 42% to 65% in Fall and Spring terms. No sustained decline of the kind critics of direct placement predicted appeared. Success rose within every racial and ethnic group in point estimates, and clearly for Latinx and Black students, while the gap between White and Latinx students barely moved, from 25 to 24 points; the data cannot rule out a substantial narrowing. These findings come from one instructor's sections and are associations, not proof that the reforms caused the change.

The takeaway is simple. Opening the gate to transfer-level statistics did not lower the share of students who passed; a larger share passed. Whether it closed the gap, one course cannot tell, and the gap did not visibly close. Colleges that want equity need equity goals of their own, complete records to track them, and results reviewed by group every term.

Statements

Use of AI. See Data and Methods, Software and AI use.

Data availability. The analysis code and the term-level totals used for grades and corequisite enrollment will be deposited at the Open Science Framework, together with the analytic dataset in a coarsened form (policy period rather than term, race collapsed to five groups, gender omitted), because at full resolution many cells contain a single student. The public files will reproduce every result except those that need race or gender by term (Model 3, the gender results, the term-by-term gap range, Table 4, Panel B, and online supplement Tables S3 to S5).

Ethics. See Data and Methods. The college was not conducting full institutional review board reviews at the time; the analysis proceeded as educational research with the department chair's approval, and the records were released without names by the Office of Institutional Research; student identifiers were used only for counts. No review board approval or exemption was obtained.

Competing interests. The author was the instructor of record for all sections analyzed and assigned the grades that make up the outcome. No financial or other competing interests.

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Appendix: Supporting Tables

Table A1 repeats the main results with and without the three COVID-19 terms. Table A2 gives the policy-period contrasts and the segmented model under three kinds of standard errors. Table A3 measures each group's gap against all students and against all other students. The online supplement gives the methods in detail and further tables (it repeats these three as Tables S1, S2, and S6).

Table A1: The main results with and without the three COVID-19 terms.

	COVID-19 terms excluded	COVID-19 terms included
Spearman ρ, term order and success rate	0.70 ($p = .005$)	0.61 ($p = .009$)
Odds ratio per year (Model 1)	1.26 [1.15, 1.37]	1.21 [1.09, 1.34]
Odds ratio per year, adjusted for term type (Model 2)	1.22 [1.13, 1.32]	1.17 [1.06, 1.29]
Odds ratio per year, adjusted for term type and race (Model 3)	1.27 [1.16, 1.40]	1.22 [1.08, 1.37]
AB 705 period vs Pre-AB 705 (Model 4)	0.64 (one term; Fisher $p = .204$)[a]	1.33 ($p = .254$)
Post-AB 705 vs Pre-AB 705 (Model 4)	1.62 ($p = .067$)	1.62 ($p = .067$)
AB 1705 vs Pre-AB 705 (Model 4)	2.61 ($p < .001$)	2.61 ($p < .001$)

Models 1 to 3: standard errors clustered by term. Model 4: conventional standard errors, adjusted for term type. [a] With the COVID-19 terms removed, the AB 705 period is the single term Fall 2019 ($n = 84$); its p value is from Fisher's exact test against the Pre-AB 705 rate.

Table A2: Policy-period contrasts and the segmented model under three kinds of standard errors: odds ratios [95% CI]; p.

	Conventional	Clustered by term	Term level (quasi-binomial)
Model 4: policy periods and term type, all 17 terms			
AB 705 + COVID-19 (vs Pre-AB 705)	1.33 [0.81, 2.18]; .254	1.33 [0.69, 2.58]; .397	1.33 [0.52, 3.41]; .551
Post-AB 705 (vs Pre-AB 705)	1.62 [0.97, 2.71]; .067	1.62 [1.06, 2.48]; .027	1.62 [0.60, 4.35]; .338
AB 1705 (vs Pre-AB 705)	2.61 [1.57, 4.33]; < .001	2.61 [1.83, 3.72]; < .001	2.61 [0.99, 6.88]; .053
Intersession term (vs Fall or Spring)	2.08 [1.32, 3.29]; .002	2.08 [1.20, 3.63]; .009	2.08 [0.87, 5.00]; .100
Segmented model: trend, term type, and a step at each policy boundary, 14 non-COVID-19 terms			
Year (per calendar year)	1.30 [1.09, 1.55]; .004	1.30 [1.11, 1.52]; < .001	1.30 [1.01, 1.69]; .046
Step at AB 705 (Fall 2019)	0.65 [0.34, 1.26]; .203	0.65 [0.39, 1.11]; .113	0.65 [0.25, 1.70]; .382
Step at AB 1705 (Fall 2023)	0.87 [0.45, 1.70]; .689	0.87 [0.43, 1.79]; .711	0.87 [0.33, 2.30]; .784
Intersession term (vs Fall or Spring)	1.95 [1.21, 3.14]; .006	1.95 [1.06, 3.58]; .032	1.95 [0.97, 3.90]; .060

Conventional: standard logistic regression. Clustered by term: cluster-robust standard errors, unreliable when a category holds one or two terms. Term level: a binomial model on term totals with the dispersion estimated from the Pearson statistic, which widens intervals to allow for swings from term to term. The segmented model's steps compare each period's level with the trend line, so a step near 1 means no jump at that boundary.

Table A3: Gaps measured against all students and against all other students, before and under AB 1705.

	Pre-AB 1705				AB 1705			
Group	n	Success, %	Gap to all	Gap to others	n	Success, %	Gap to all	Gap to others
White	137	68.6	+17.1	+22.4	32	84.4	+16.2	+19.8
Asian	46	60.9	+9.3	+10.1	11	72.7	+4.5	+4.8
Latinx	287	43.6	−8.0	−15.9	97	60.8	−7.4	−16.8
Black	49	26.5	−25.0	−27.3	13	61.5	−6.7	−7.2
All students with race recorded	578	51.6			173	68.2		

Enrollments with race recorded, all terms. Gap to all: the group's success rate minus the rate for all students with race recorded. Gap to others: the group's rate minus the rate for all other students with race recorded (percentage point gap minus one). Percentage points; negative means below the reference. The four groups do not add to the total, which also includes other and unknown race.